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Paleomagnetic study of Permian and Mesozoic sedimentary rocks from Northern Thailand supports the extrusion model for Indochina

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ABSTRACT

The mode of deformation of southeast Asia resulting from the Indian collision is still controversial: is convergence between stable India and Siberia absorbed principally by crustal shortening and thickening and minor strike-slip faulting or by a mechanism of extrusion involving crustal thickening and lateral escape of blocks along major strike-slip zones? In order to test these models, we have carried out a paleomagnetic study of sedimentary rocks of Jurassic to Cretaceous age on the Khorat Plateau in Thailand. A large discrepancy between the poles of Indochina presented in this paper and a compilation of South China block pole positions both at the Triassic/Jurassic and the Jurassic/Cretaceous boundary imply some 1500 ± 800 km of post-middle Cretaceous left-lateral slip along the Red River and Xian Shui He fault zones and approximately $14 \pm 7°$ of clockwise rotation for the Indochina block relative to the South China block. Such relative motions are in agreement with a model of lateral extrusion of Indochina during the India–Asia collision. Additional data of Permian, Upper Triassic and Lower Jurassic age collected in Indochina indicate a post-late Permian remagnetization for the upper Permian limestones. A comparison of these data with those from Yunnan (South China), the North China block and the South China block shows that these blocks have probably been in contact at least since the late Triassic.

1. Introduction

The India–Asia collision has significantly affected the Cretaceous landscape of central and southeastern Asia since the beginning of the Tertiary (Fig. 1a). As a result, the geological history of the assembly of the different blocks involved in the pre-Tertiary collision is still a matter of heated debate. The Indochina block, for example, is considered to have collided with the South China block in the Tournaisian–Visean [1], the late Triassic [2–4], or during the Jurassic to early Cretaceous [5]. The affinities of the late Triassic vertebrates between Indochina and Laurasia (Eurasian–North American continent) led Buffetaut [6] to propose that the Indochina block had been sutured to the SCB at least since that period, or that continental connections between Indochina, the South and North China blocks and Siberia existed at that time. Moreover, ancient structures were severely deformed or masked during the successive collisions, especially during the collision and subsequent penetration of the Indian plate into Eurasia between 60 and 50 Ma [7].

Models of continental shortening and deformation of these regions after the onset of the collision are also strongly debated. The convergence between stable India and Eurasia (or Siberia), which is in excess of 2600 km [7–10], is, according to England and Houseman [11], absorbed principally by crustal shortening and thickening and minor strike-slip faulting, while Tapponnier and Molnar [12], Tapponnier et al. [10] and Peltzer and Tapponnier [13] argue for a mechanism involving crustal thickening and lateral escape of blocks along major strike-slip zones (the extrusion model). In this controversial context, paleomagnetic studies can provide an independent tool to constrain the paleogeographic history and the deformation mode of these blocks.

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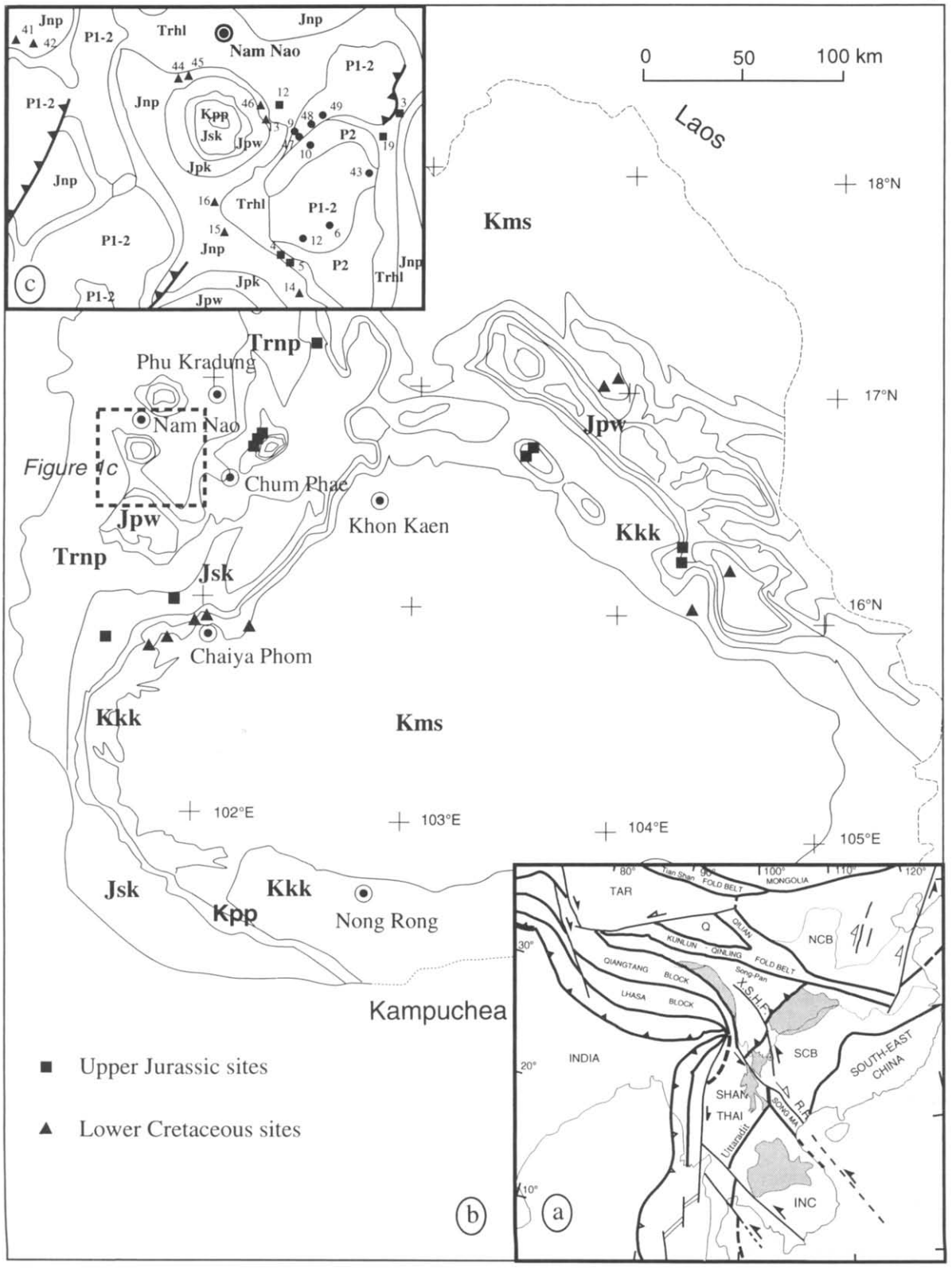


Fig. 1. (a) Sketch map of the tectonic elements of eastern Asia, modified from Yang et al. [61] and Peltzer and Tapponnier [13]. The major blocks are: *TAR* = Tarim; *NCB* = North China block; *SCB* = South China block; *Q* = Qaidam; *IND* = India; *INC* = Indochina. The Sichuan, central Yunnan, Markam and Khorat basins are shaded. Arrows correspond to the sense of strike-slip motion on major faults (open for recent, closed for older phases). (b) Simplified geological map of the Khorat basin in Thailand (based on the geological map of Thailand; 1/50000 [39]) with sampling sites indicated. (c) Simplified geological map of the western Khorat basin in the Chum Phae area with sampling sites indicated [after 39]. Dots, squares and triangles denoted the sampling sites of the Permian, uppermost Triassic and lowermost Jurassic respectively.

The paleomagnetic database from the North China block (NCB), the South China block (SCB) and the Central Asia blocks (Dzunggar, Tarim and Tibet) has improved tremendously over the last decade [14–21] and now provides both a coherent first-order picture of the eastern Tethys paleogeographic history and an indication of how continental shortening was absorbed within the southern part of Eurasia. Among these blocks, Indochina suffers an important lack of good-quality paleomagnetic data, as was pointed out by Haile and Briden [22]. Moreover, some additional uncertainties regarding the primary nature of existing poles [23,24] preclude any integration of this block into the southeastern Asian realm.

2. Geological setting

The Khorat plateau is a part of Sundaland, a continental block which probably became consolidated during the early or middle Mesozoic. Sundaland includes Indochina, the Sunda shelf, Peninsular Malaysia and Sumatra, southwest Borneo, a part of Burma, a part of Yunnan south of the Red River fault and, possibly, parts of southwest Tibet (Fig. 1a). Its present-day boundaries are marked by ancient suture zones to the northwest (Nan–Uttaradit suture) and northeast (Song Ma suture) of possible upper Triassic age. A large variation in the ages of the associated granites (from Triassic to upper Jurassic) underlines our poor understanding of the geological configuration in the early–middle Mesozoic. Active Tertiary faults, the Red River fault to the northeast and the Wang Chao–Three Pagodas fault system to the southwest, also mark boundaries the area. Along the length of the Wang Chao and Three Pagodas faults there is a N–S trending belt of granites which represent the probable continuation of the Uttaradit suture zone. Although the ages are not well constrained [25,26], these granites are easy to recognize and mark the boundary of Indochina with respect to the Shan–Thai–Malay block. After the major pre-late Triassic tectonic phase known as the Indosinian orogeny, the Khorat became a basin filled by a thick continental succession of molasse (mainly redbeds) ranging in age from upper Triassic (Norian) to upper Cretaceous. In the region of the west Khorat basin, the Indosinian orogeny is marked by the unconformity of the Mesozoic succession over a strongly folded marine limestone succession of upper Permian age, which has been well dated by a rich *Fusulina* fauna [27,28]. Large continental basins of the same age can also be found in central Yunnan and Sichuan (Fig. 1a).

The Khorat Group is usually divided into several formations based on lithostratigraphic cycles. However, sparse fossils (mostly wood and plant material) and poor exposure have led to difficulties in dating, dividing and establishing a full sequence. Possible unconformities are also difficult to establish. Recent progress in vertebrate paleontology has led to more precise definition of the stratigraphic ages, however [29]. The lowermost formation of the Mesozoic Khorat Group, the Huai Hin Lat Formation, consists of a succession of limestones and sandstones and contains a rich freshwater vertebrate fauna (e.g., fishes, stegocephalians, turtles and phytosaura). The assigned age is Norian (late Triassic). This succession is conformably overlain by a red siltstone formation (the Nam Phong Formation), which has been assigned a post-Norian–pre-middle Jurassic age. This succession is overlain by the Phu Kradung Formation, which includes various vertebrates, such as the crocodilian *Sunosuchus thailandicus* of middle Jurassic age [6]. (Such fauna are also found in formations of the same age in the Kansu province (North China block) [30].) The overlying succession consists mainly of sandstones that are divided into the Phra Wihan Formation (middle Jurassic), the Sao Khua Formation (late Jurassic) and the Phu Phan and Khok Kruat formations (early to middle Cretaceous on the basis of the rich vertebrate fauna). The early Cretaceous dinosaurs *Psittacosaurus* and *Thaiodus ruchae* found in the Khok Kruat Formation can be compared with those found in coeval strata from the Aptian–Albian of South Tibet (Lhasa block) [29,31].

3. Previous paleomagnetic studies on the Khorat plateau

The paleomagnetic data on the Indochina block and Shan–Thai block obtained in the 1970's and the beginning of 1980's are in general of poor quality [32–34]. Indeed, although some thermal demagnetizations in pilot studies were carried out

up to 600°C, most of the samples were routinely demagnetized to temperatures never exceeding 200–300°C, a rather low temperature for separating components often carried by hematite and sometimes by both magnetite and hematite. Moreover, most of these results were based on a small number of sites or samples. More recently, a preliminary paleomagnetic study of upper Triassic rocks from the western Khorat basin (Northeast Thailand) was carried out by Achache and Courtillot [23], and provided an upper Triassic pole for the Khorat plateau. Both a positive fold test and a reversed polarity site seemed to confirm the primary nature of the magnetization. However, this study was based on a few sites only. A recent reappraisal of these data, along with a new analysis of additional Devonian to Permian samples and one Jurassic site, led Chen and Courtillot [24] to suggest that most of the Indochina block may have suffered a complete remagnetization after the Triassic, and possibly after the Cretaceous or early Cenozoic. Further sampling of the sequence was thus necessary in order to test the hypothesis of a general remagnetization linked to the India–Asia collision in the Cenozoic. Even if a complete overprint did occur, the process and age of remagnetization needed to be ascertained. We therefore collected additional Permian and Triassic samples, as well as Jurassic and lower Cretaceous samples, on the Khorat plateau (Fig. 1b and c). These new data are first described and analyzed, and then merged with earlier data.

4. Paleomagnetic and rock magnetic study

The paleomagnetic sampling was carried out using a portable petrol-powered drill. The natural remanent magnetizations (NRM) were measured using a CTF cryogenic magnetometer in the shielded room at the IPG in Paris. Fifteen specimens were thermally demagnetized and then measured for comparison using a JR4 spinner magnetometer in the Saint-Maur laboratory. The stepwise thermal and alternating field (AF) demagnetizations were carried out up to 700°C and 100 mT with a laboratory-built oven and a Schonstedt GSD1 single-axis demagnetizer. A mixed thermal and AF demagnetization was sometimes used for some rocks containing magnetite as a

TABLE 1

Characteristic remanent magnetization direction for the upper Permian from the Chum Phae area of the Indochina block (sampling location 16.67°N, 101.83°E)

Site	rock type	Strike/Dip	N	Dg	Ig	Ds	Is	κ	α95
Th6(*)	Limestone	28°/52°	9	25.5	45.4	66.1	30.0	69.0	6.2
Th9(*)	Limestone	35°/42°	2	32.2	27.6	54.0	25.1		
Th10	Limestone	50°/80°**	7	193.4	-22.7	257.3	-38.6	16.1	15.5
Th12(*)	Limestone	34°/27°	8	24.3	40.5	42.8	38.9	178.0	4.2
Th31	Limestone	250°/25°	5	32.4	42.9	19.9	25.3	129.5	6.7
Th43	Limestone	320°/43°	6	4.3	53.5	23.4	17.9	36.6	11.2
Th47	Limestone	30°/45°	7	26.6	40.1	59.0	29.1	115.9	5.6
Th48	Recrystallized Limestone	30°/42°	8	39.8	38.0	63.3	21.6	28.9	10.5
Th49	Recrystallized Limestone	55°/43°	6	21.6	36.8	62.3	47.6	206.1	4.7
Mean of normal sites			8	26.6	41.0	48.4	30.6	18.9	13.1
Mean of all sites			9	24.8	39.0	51.4	31.8	17.6	12.6
Fold test				kg/ks=2.5 > F(16,16)=2.33 indicates negative fold test at 95% confidence level					
Progressive unfolding				at 30% tilt correction k(30%)/k(100%)=4.27 D= 33.0°, I=39.4° , k (30%)=75.4, α_{95}=6.0°					

* From [24].

** Indicates the strike/dip only in the upper part of the site, which differs form the overturned, lower part where the bedding attitude is 50°/72°NW.

remanence carrier. The bulk magnetic susceptibility of the specimens was measured at each demagnetization step in order to monitor magnetic mineral transformations during progressive heating. The magnetic directions were analyzed by principal component analysis [35], and the site-mean direction was calculated using McFadden and McElhinny's statistics [36]. Determination of minerals of the magnetic fraction was attempted on selected polished thin sections of limestones and red sandstones, and under both the optical and the scanning electronic microscope (SEM). The chemical composition was determined using attached energy dispersion X-ray analyses and microprobe analyses. The determination of Curie temperatures was also carried out on the separated magnetic minerals with a horizontal balance in air. The rock was first crushed into millimetre-size chips and then bathed in a buffered (pH = 4.5) acetic acid solution for 2 days. The chips were then washed with distilled water. Finally, the magnetic minerals, attracted

TABLE 2

Characteristic remanent magnetization directions for the upper Triassic and lower Jurassic of the Indochina block (sampling location 16.67°N, 101.83°E)

Site	rock type	Strike/Dip	Temp. range	N	Dg	Ig	Ds	Is	κ	α95/A95	λ	ϕ
Upper Triassic Huai Hin Lat formation												
Th12	Limestone	152/36	300°-510°	8	231.0	-13.9	227.1	-43.2	23.4	11.7	45.4	172.7
3*	Limestone	103/10	300°-500°	13	36.7	46.2	41.6	54.6	70.0	5.0	48.7	157.1
4*	Sandstone	105/12	300°-510°	12	32.8	31.5	35.6	41.8	11.0	13.6	55.9	173.3
5*	Sandstone	110/20	300°-510°	14	32.5	29.7	36.6	49.1	78.0	4.5	54.0	163.4
19*	Limestone	20/13	300°-510°	9	29.1	35.9	37.0	33.1	59.0	6.7	54.7	184.0
Mean	of all sites			5	36.9	31.7	-	-	33.8	13.3		
				5	-	-	39.5	44.4	85.0	8.3		
Mean	of VGPs								109.4	7.3	52.1	169.8
Lower Jurassic Nam-Phong formation												
Th 13	Siltstone	160°/40°	560°-690°	9	232.7	-2.3	226.7	-40.7	20.6	11.8	45.8	175.4
Th 14**	Siltstone	110°/25°	NRM-540°	7	203.2	-9.3	203.8	-33.8	48.8	16.4	67.3	183.5
Th 15**	Sandstone	110°/3°	NRM-540°	5	197.7	-35.1	197.6	-38.0	16.2	16.1	72.7	173.4
Th 16	Siltstone	240°/8	500°-690°	8	53.6	39.1	47.3	37.7	42.0	8.6	45.2	178.4
Th 42	Siltstone	280°/16°	600°-690°	7	55.4	55.4	43.1	42.4	30.6	11.4	49.1	173.4
			300°-550°	7	36.5	46.4	31.0	31.1	187.5	4.4	60.3	187.1
Th 44	Siltstone	81°/32°	120°-550°	8	32.6	14.2	44.0	36.3	165.6	4.3	48.2	180.0
Th 45	Sanstone	100°/21°	300°-500°	6	32.7	20.7	38.0	39.7	68.8	8.1	53.8	176.3
Th 46	Sandstone	168°/30°	300°-500°	4	50.6	22.9	38.4	48.2	25.2	18.7	52.7	165.2
Mean of normal sites				5	43.6	30.7	42.3	40.9	212.2	5.3		
Mean of reversal sites				3	212.0	-16.0	209.0	-38.1	42.5	19.1		
Mean of all sites				8	39.0	25.4	-	-	14.0	15.3		
				8	-	-	37.2	40.1	71.6	6.6		
Mean of VGPs				8					58.0	7.3	54.4	175.6
Mean of Upper Triassic/Lower Jurassic				13	38.2	27.9	-	-	18.8	9.8		
				13	-	-	38.1	41.8	77.5	4.7		
Mean of VGPs				13					72.7	4.9	53.6	173.3

Fold tests

1. Mean of Huai Hin Lat formation — kg/ks=2.51

1. Higher temperature components of Nam-Phon Formation — ks/kg=5.1 > F(14,14)=3.7
indicate positive fold test at 99% confidence level

3. Mean of Upper Triassic/Lower Jurassic — kg/ks=4.12>F(24,24)=2.66
indicate positive fold test at 99% confidence level

* Sites reported by Chen and Courtillot [24].

** Mean results obtained by intersection of great circles with directly observed directions [36].

by a rare earth magnet, were collected. The paleomagnetic and rock magnetic results are described below, going from the oldest to the youngest formation, and are summarized in Tables 1, 2 and 3. A more detailed presentation of these data is included in Yang [37].

5. Upper Permian results

Two main magnetic components in the Permian limestones can be separated both by thermal and AF demagnetization (Fig. 2). The lower temperature component direction ($D = 358.8°$, I

TABLE 3

Characteristic remanent magnetization directions for the upper Jurassic and lower Cretaceous of the Khorat basin

Site	Location	Strike /Dip	Component	N	Dg	Ig	Ds	Is	k	α95 /A95	λ	φ
Lower Cretaceous results												
21	16°57'/103°57'	240°/2°	B	12	24.5	42.0	23.1	40.8	474.6	2.0	67.4	173.9
21			C	12	28.6	39.0	27.2	37.9	182.2	3.2	64.0	180.2
22	16°57'/103°57'	25°/5°	B	8	25.4	37.9	29.3	37.7	39.7	5.8	62.0	180.8
22			C	6	31.6	31.7	34.5	30.9	97.3	6.5	57.0	189.4
23	16°56'/103°57'	162°/5°	B	7	30.8	39.1	28.3	42.3	67.1	7.4	62.6	173.5
23			C	8	32.3	31.1	30.5	34.4	114.3	5.2	61.0	185.5
28	16°00'/102°31'	350°/3°	B	13	21.0	40.2	23.1	38.6	221.2	2.8	67.4	174.1
28			C	13	19.9	38.5	21.9	37.0	151.1	3.4	68.7	176.6
33	15°48'/102°13'	130°/5°	B	10	26.9	37.8	24.6	42.2	250.1	3.1	65.4	176.8
33			C	10	24.5	36.1	22.2	40.3	191.1	3.5	67.9	169.8
34	15°51'/101°58'	75°/11°	B	6	22.3	35.9	28.4	44.4	315.3	3.8	61.7	166.1
34			C	5	25.3	35.0	31.6	43.2	100.0	7.9	59.1	169.3
35	15°51'/101°58'	68°/12°	B	8	26.2	31.9	33.0	39.4	72.0	6.6	58.2	175.0
35			C	7	32.5	34.9	40.6	41.2	94.2	6.3	51.1	173.8
37	15°48'/101°50'	65°/8°	B	5	26.9	39.3	32.7	43.9	313.1	4.3	58.0	168.4
37			C	5	39.5	34.5	44.9	37.6	102.2	7.6	47.2	177.7
38	15°48'/101°48'	0/0	B	12	31.2	34.7	31.2	34.7	192.1	3.1	60.1	180.8
38			C	6	35.6	36.3	35.6	36.3	107.5	6.5	55.9	179.0
39	15°48'101°41'	90°/8°	B	7	25.0	33.2	27.4	40.2	88.1	6.5	63.2	172.0
39			C	5	27.5	34.3	30.4	41.4	73.6	9.0	60.4	171.2
Mean			B	10	26.0	37.2	-	-	388.5	2.5		
Mean			B	10	-	-	28.1	40.5	398.5	2.4		
Mean	of VGPs		B	10					390.6	2.4	62.7	173.3
Mean			C	10	29.8	35.3	-	-	232.9	3.2		
Mean			C	10	-	-	32.0	38.2	140.8	4.1		
Mean	of VGPs		C						113.9	4.5	59.4	177.4
Upper Jurassic results												
17	16°40'/102°14'	330°/7°	C	8	25.2	39.4	28.1	33.6	180.4	4.1	63.2	184.3
18	16°40'/102°14'	340°/8°	C	8	21.5	37.1	25.4	31.6	115.6	5.3	65.7	187.5
19	16°40'/102°14'	335°/8°	C	6	20.1	39.5	24.2	33.7	320.0	3.8	70.2	172.2
20	17°14'/102°28'	295°/5°	C	8	205.3	-44.8	205.3	-39.8	87.6	6.3	65.6	175.5
25	16°19'/104°19'	177°/8°	C	10	31.0	32.2	26.3	36.4	194.8	3.5	64.7	181.3
26	16°19'/104°19'	177°/8°	C	7	29.2	36.9	23.6	40.8	133.9	5.3	66.7	172.9
30	16°43'/103°33'	313°/7°	C	8	24.8	40.6	26.4	34.0	189.7	4.0	64.8	185.2
31	16°43'/103°30'	120°/3°	C	5	31.8	37.3	31.9	40.8	117.9	7.1	59.4	175.7
32	16°43'/103°30'	173°/11°	C	7	36.2	36.3	29.1	43.4	295.9	3.5	61.6	171.0
40	15°46'/101°33'	100°/3°	C	6	25.3	35.8	25.9	38.7	309.6	3.8	65.0	175.5
Mean			C	10	27.1	38.1	-	-	247.4	3.1		
Mean			C	10	-	-	26.6	37.3	339.2	2.6		
Mean	of VGPs			10					443.0	2.3	64.8	178.1
Mean	J+K		C+B	20	26.6	37.7	-	-	313.3	1.8		
	J+K		C+B	20	-	-	27.3	38.9	328.3	1.8		
Mean	of VGPs		C+B	20					377.9	1.7	63.8	175.6
Mean	J+K		C+C	20	28.5	36.7	-	-	224.7	2.2		
Mean	J+K		C+C	20	-	-	29.3	37.8	181.6	2.4		
Mean	of VGPs		C+C	20					156.2	2.6	62.1	177.8

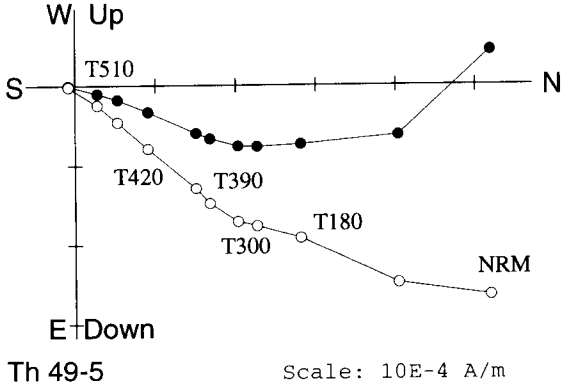
B and C are medium temperature component (200–550°C) and higher temperature component (550–670°C) respectively.

$= 28.6°$, $\alpha_{95} = 4.0$, $N = 37$, in geographic coordinates) is removed at 300°C and is statistically identical to the local present dipole field direction ($I = 32°$). The fact that the precision parameter (k) decreases gradually from 36.3 to 7.1 during unfolding indicates that this component clearly post-dates the folding. The higher temperature component decays to the origin from 300°C to around 510°C (Fig. 2). Two components are also present in AF demagnetization (not shown). Two polarities are observed, with a north-northeastly declination (downward inclination) and a nearly antipodal south-southwestly direction (upward) before bedding correction. The unblocking temperature range, along with the low coercivity field (< 0.3 T) in the IRM acquisition curves (not shown), probably indicate some member of the titanomagnetite family as a principal carrier of the remanence in these rocks. Maghemite or low blocking temperature titanomagnetite may also be present at site TA31, with an abrupt decrease in magnetization between 300 and 380°C and then a reversed higher temperature component revealed between 380 and 510°C. The mean directions for the high temperature component are given before and after bedding correction in Fig. 3 and Table 1. The dispersion of the ChRM direction clearly increases after tectonic tilt correction, and the ratio of k_g/k_s for nine sites gives a negative fold test at the 95% level using the McElhinny fold test [38]. However, progressive unfolding reveals a maximum value of the precision parameter (k) at the 30% unfolding level (Fig. 3c). The ChRM direction may therefore be a remagnetization acquired during folding. Mean directions of the ChRM at 30% unfolding are computed at the site level ($D = 33.0°$, $I = 39.4°$, $\alpha_{95} = 6.0$), giving a paleopole at $\lambda = 58.4°$N, $\phi = 176.2°$E, $d_p = 4.3°$, $d_m = 7.2°$ (Table 1). The Triassic age of this tectonic phase is fairly precisely known and can be related to the Indosinian orogeny. A cross section along the Lom Sak–Chum Phae road (location of our sampling area) clearly shows the vertical and strongly deformed marine Upper Permian units unconformably overlain by the upper Triassic layers of continental facies. The discordance is also clearly shown on the 1/25000 geological map of Thailand (Changwat Phetchabun sheet [39]).

6. Upper Triassic–lower Jurassic results

6.1 Huai Hin Lat Formation (upper Triassic)

The Huai Hin Lat Formation has already been investigated by Achache and Courtillot [23] and Chen and Courtillot [24], and the conclusions



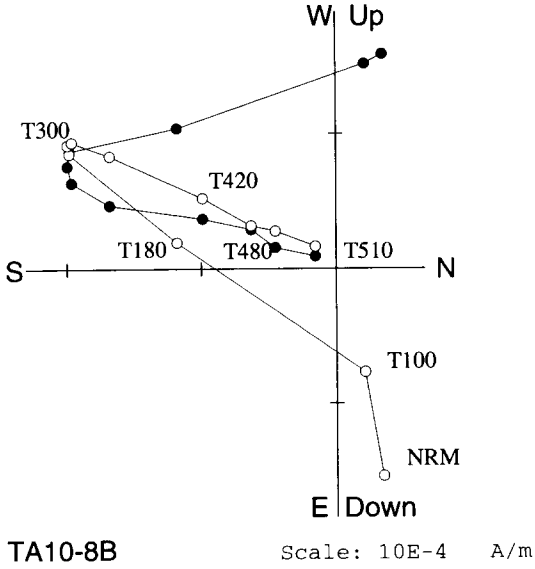


Fig. 2. Orthogonal vector diagrams of representative specimens from the Permian limestones of the western Khorat basin. Closed symbols refer to the projection on the horizontal plane in geographic coordinates (open symbols for the vertical plane).

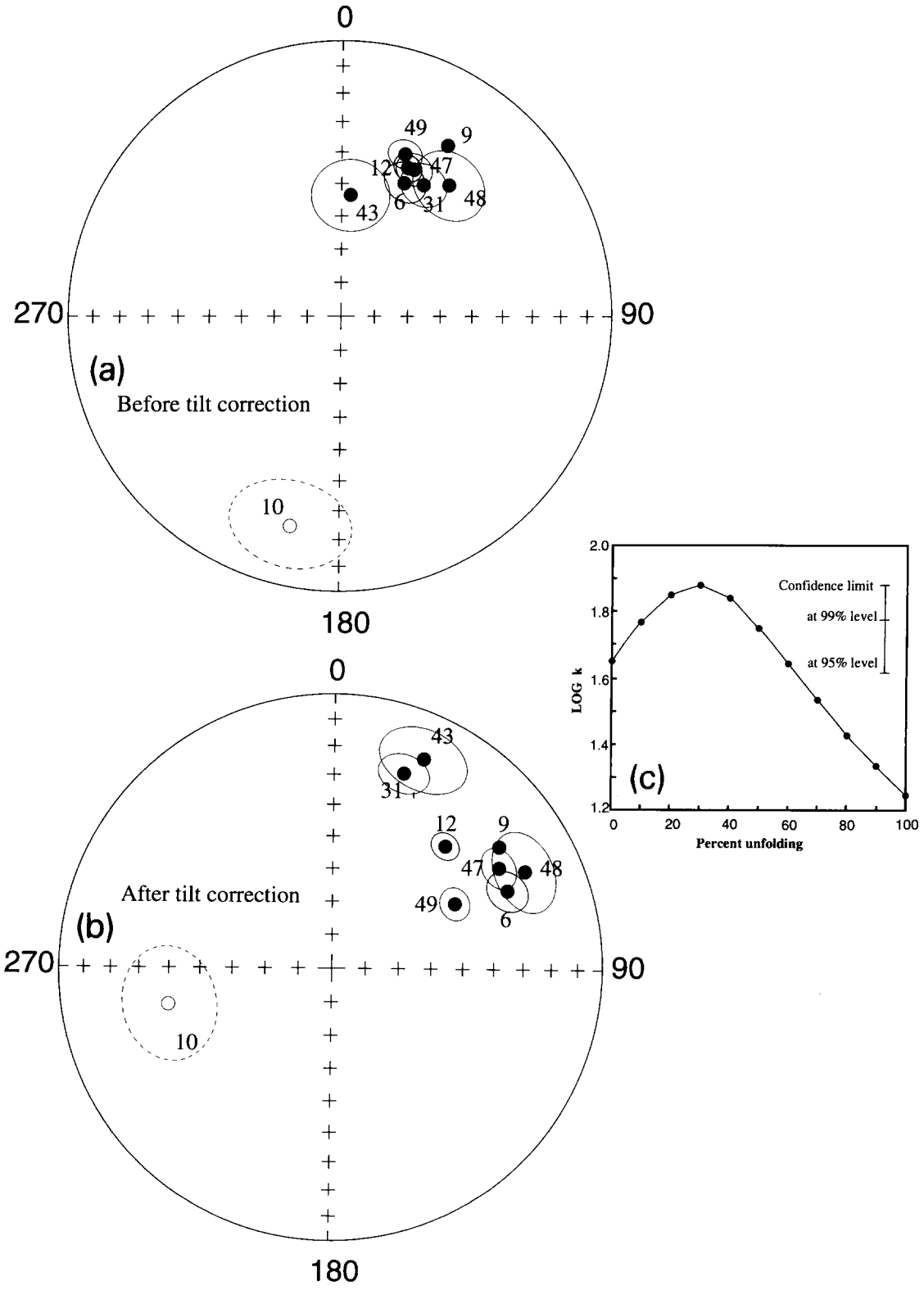


Fig. 3. (a and b) Equal-area projection of site-mean direction of the Permian of the western Khorat basin. Closed symbols represent directions plotted onto the lower hemisphere, open symbols onto the upper henisphere. Circles are the 95% confidence limits around the site-mean directions. (c) Progressive unfolding of the site-mean directions reveals synfolding magnetization, with a peak in the precision parameter k at 30% unfolding (95% and 99% confidence limits shown).

from these two papers are at variance with each other, both for directions and fold test significance. The origin of this variance lies in a different interpretation of the ChRM component of site 8. This site shows a reversed polarity, but the ChRM directions are badly defined due to a strong mineralogical transition during thermal demagnetization. Moreover, alternating field demagnetization alone is unfortunately insufficient to reveal the characteristic component (see [23], Fig. 3). We therefore resampled a lateral equivalent of Achache and Courtillot and Chen and Courtillot's site 8 in the Chum Phae area (site Th12 in Fig. 1c), just a few metres from the previous site. An isothermal remanent magnetization (IRM) acquisition curve shows that magnetic saturation is reached slightly after 0.2 T (Fig. 4a). A slight increase in the susceptibility above 350°C (Fig. 4b) indicates a probable transformation of some of the magnetic minerals during the heating treatments [40,41]. This transformation may explain the difficulty of characterizing the final

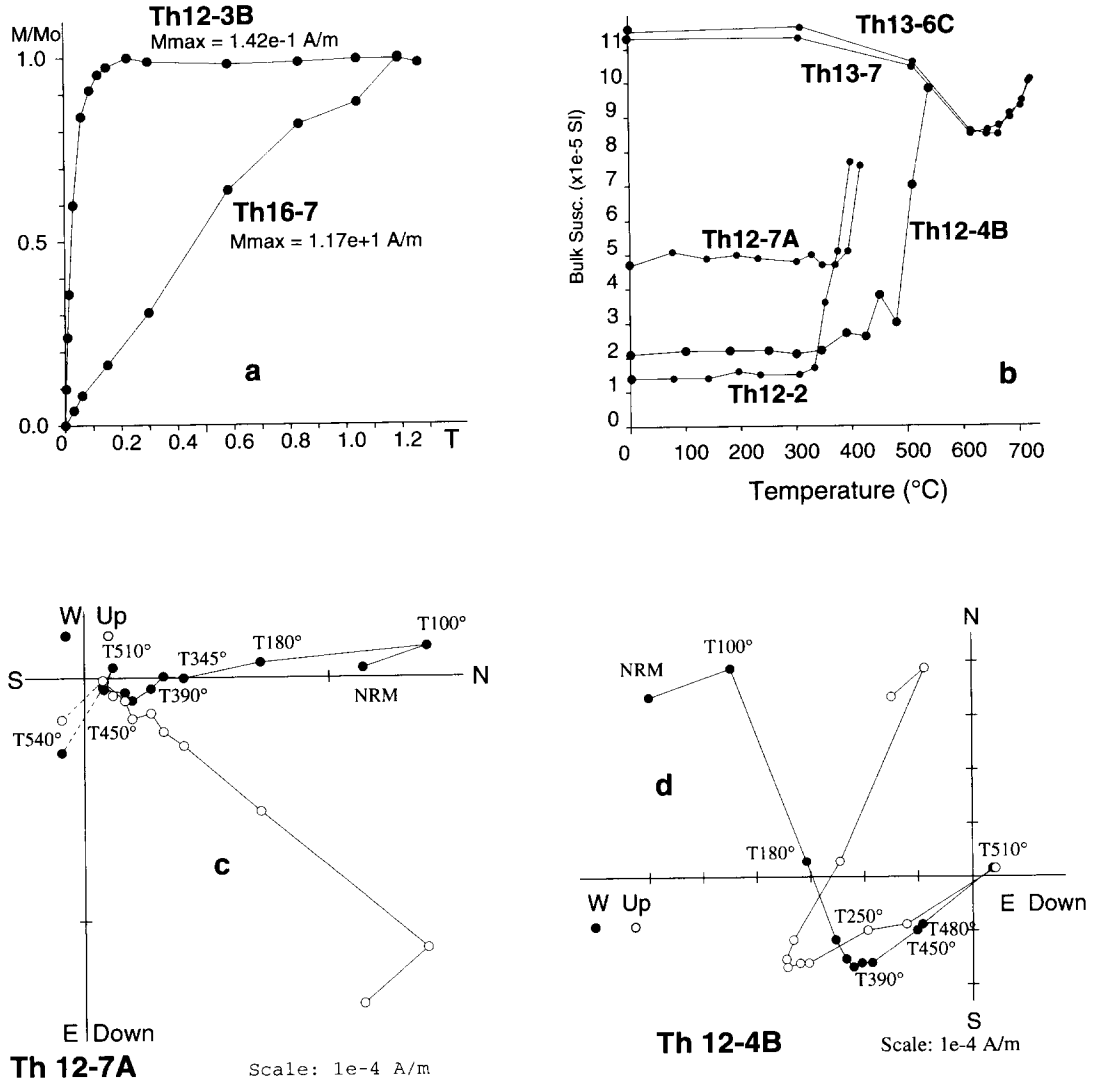
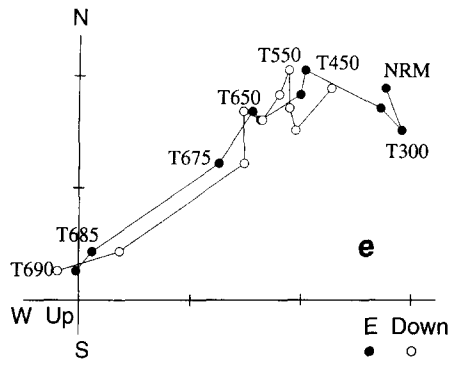
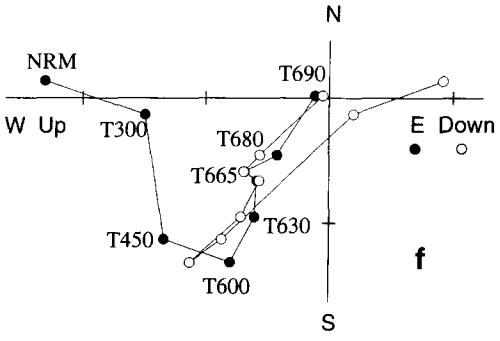


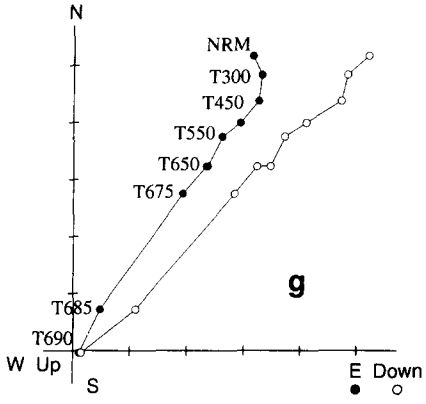
Fig. 4. (a) IRM acquisition curve (limestone of the Huai Hin Lat formation, Th12-3B; sandstone of Nam Phong formation, Th16-7). (b) Changes in bulk susceptibility of the resampled upper Triassic site and lower Jurassic sandstones during heating. (c and d) Orthogonal vector diagrams for the resampled upper Triassic specimens from the Huai Hin Lat Formation (upper Norian).



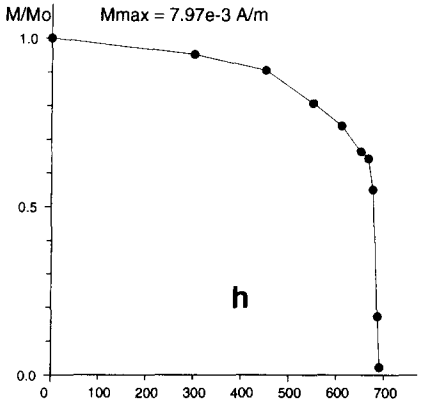
Th13-8 Scale: 1e-3 A/m



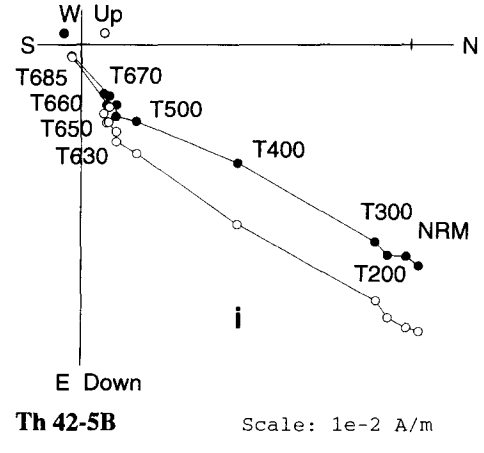
Th 13-5 Scale: 10E-3 A/m



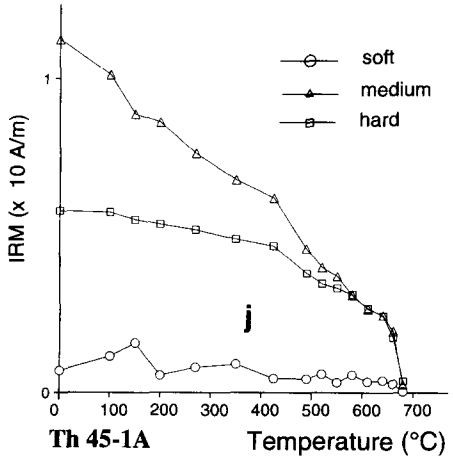
Th16-5B Scale: 10E-3 A/m



Th16-5B Temperature (°C)



Th 42-5B Scale: 1e-2 A/m



Th 45-1A Temperature (°C)

Fig. 4. (e, f, g and i) Orthogonal vector diagrams for representative specimens from the Nam Phong Formation (lower Jurassic) during thermal demagnetization, in stratigraphic coordinates. (h) Normalized decay curve of the magnetization intensity during heating. (j) Thermal demagnetization of a triaxial differential IRM (0.12, 0.4, 1.3T; method described in Lowrie [43]).

direction of the magnetic vector. Two magnetization components are generally isolated by progressive demagnetization, as shown in the Zijderveld orthogonal vector projections [42] (Fig. 4c and d). The first component is destroyed below 350° C, whereas the second, higher temperature component shows a univectorial decay through the origin to 510°C. These characteristics indicate that some form of (titano)-magnetite is the likely carrier of the ChRM. A composite demagnetization using both thermal (to 250°C) and then alternating field techniques (up to 45 mT) gives an identical result. From the top to the bottom of the sampled section, a sequence of normal, reversed (five samples) and normal polarity is found. The new site was combined with the four sites

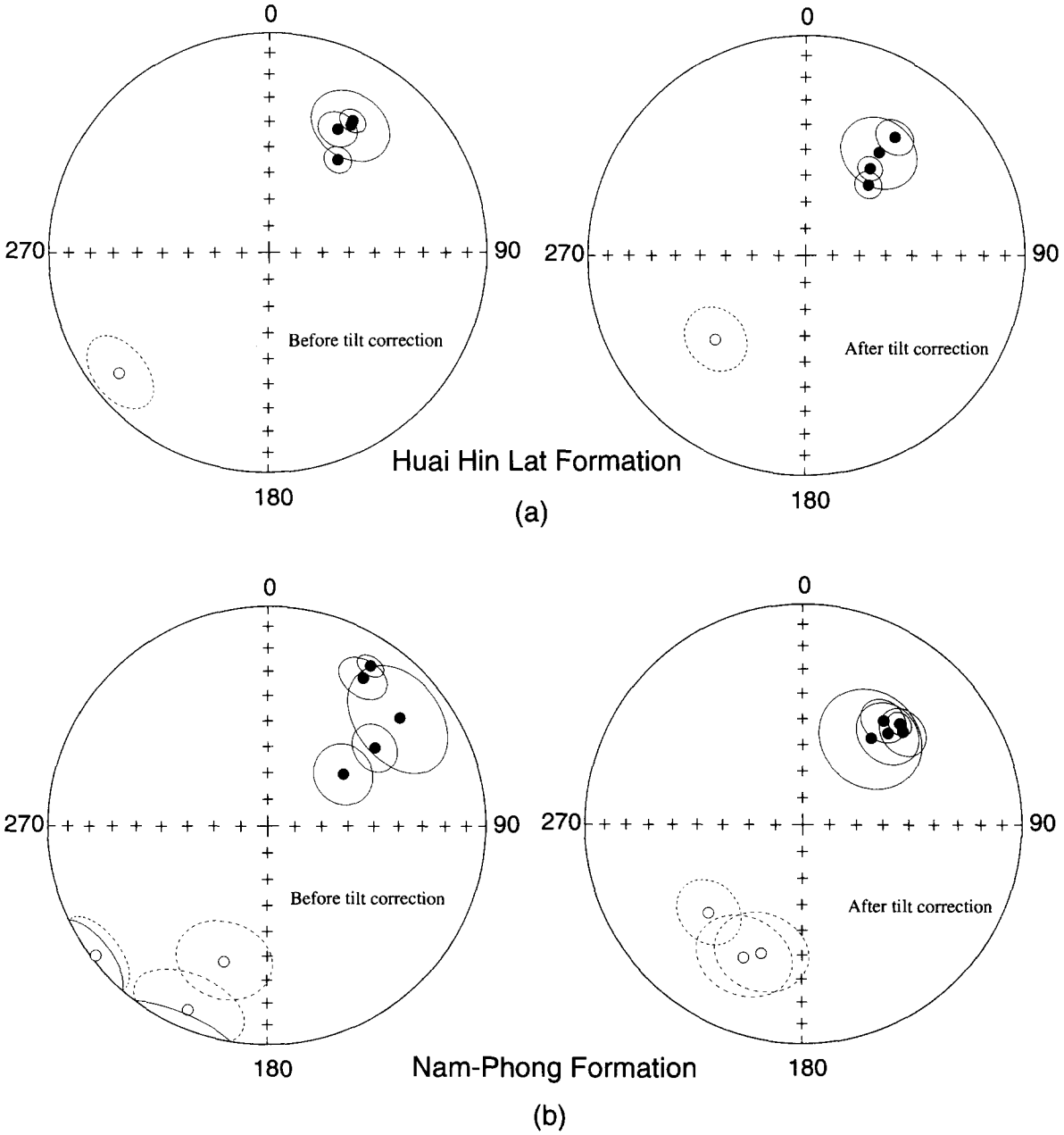


Fig. 5. Equal-area projection of the Norian site directions (a) and of the lower Jurassic sites (b) before and after tectonic correction.

from the late Triassic Huai Hin Lat Formation already studied by Chen and Courtillot [24]. Because the dating of the rhyolitic tuffs near Sara Buri City is poor, we have rejected their site 27 from the mean. The increase in precision parameter is not sufficient for a positive fold test at the 95% level of probability, although bedding correction leads to an increase in the precision parameter (from 33.8 to 85.0). We should bear in mind that the relatively minor bedding corrections are not very favourable for obtaining a significant fold test (see Fig. 5a and Table 2). A paleomagnetic pole at $\lambda = 52.1°N$, $\phi = 169.8°E$, $A95 = 7.3$, $N = 5$ is derived (Table 2). This value is statistically indistinguishable from that of Achache and Courtillot [23].

6.2 *Nam Phong Formation (lower Jurassic red sandstone)*

The general behaviour is illustrated by sites 16 and 13 shown in Fig. 4a, b and e–h. After removing a low unblocking temperature component (450°C) in the direction of the present field, the Zijderveld plots most commonly show univectorial decay toward the origin with unblocking temperatures in the range 650–680°C, as shown by the normalized intensity decay curve (Fig. 4h). Both normal and reversed antipodal directions are found at site 13 (Fig. 4e and f). The IRM acquisition curve shows that a field of 1.3 T is not sufficient to reach complete saturation (Fig. 4a, Th16-7). The hematite family is thus a possible carrier of the remanence. The susceptibility curves measured after each step of thermal demagnetization show no major evolution in the susceptibility values during the thermal treatments (Fig. 4b).

At some sites (14 and 15), a high unblocking temperature component carried by hematite partly or completely overlaps a lower temperature component, resulting in great circle demagnetization trajectories on the stereographic projection (not shown). The demagnetization trajectories do not always pass through the origin on the Zijderveld diagram, and no reliable directions could be determined above 650–690°C in 20% of the samples; these samples were therefore rejected. Mineralogical transformation during heating may be the cause of this behaviour [41]. A mean direction was calculated using both remagnetization great circles intersection and directions determined by principal component analysis [36].

Sites 42 and 45 display slightly different behaviour (Fig. 4i and j). A medium temperature range component is evident from thermal treatment between 300 and 500°C, while only a small part of the magnetization remains between 500 and 690°C. The high temperature component usually represents less than 20% of the initial intensity. Moreover, the direction at site 45 often becomes erratic and prevents a precise determination of this component (the same is true for the site 44). To characterize better the magnetic carriers, we have thermally demagnetized a composite IRM in the manner described by Lowrie [43]. On the same core, an IRM of different coercive force (1.26, 0.4 and 0.12 T) is applied successively to each one of the three axes and then thermally demagnetized. The results are shown in Fig. 4j. The demagnetization of the IRM shows that medium coercivity minerals are predominant. An inflexion is seen between 400 and 500–550°C, part of the magnetization remains until 680°C, and then a final drop occurs. The high coercivity component is also important and decreases gently until 650°C, followed by a sudden drop. In an attempt to identify the magnetic carriers, we have tried to determine their Curie points with a horizontal balance. The thermomagnetic curve shows two main inflexions, giving Curie points around 500°C and 685°C [37]. Two main magnetic carriers are thus indicated by these experiments. The first one is probably a mineral of the titanomagnetite family, with a low Curie point. The thermal demagnetization of NRM and IRM and the thermomagnetic curve all show that this mineral is dominant. Analyses of polished thin sections in reflected light show detrital iron-oxide minerals with their typical steel-grey colour, and microprobe analysis further confirms the presence of Ti. The rounded detrital magnetites show evidence of abrasive transport. The second magnetic carrier is clearly some form of hematite, as indicated by Curie temperatures and coercivity.

One site (site 41) has been rejected, due to a strong present-day earth field overprint. In this case, it was impossible to isolate any final component.

All together, eight sites out of nine revealed a characteristic direction. Three sites are of re-

versed polarity and five of normal polarity (Table 2 and Fig. 5b). The sampling sites were located on different flanks of gentle folds, giving an opportunity to judge the stability of the remanent magnetization. The fold test is positive at the 95% level ($k_s/k_g = 5.1$). Folding could have occurred at any time between the middle–late Jurassic Yanshanian orogeny and the late Tertiary. The mean direction carried by magnetite is statistically indistinguishable from that carried by hematite, suggesting that the magnetization of both minerals became locked over a short time interval (site 42).

A pole for the lower Jurassic is derived at $\lambda = 54.4°N$, $\phi = 175.6°E$, $A95 = 7.3°$ (Table 2 and Fig. 6). Figure 6 shows that our poles from the Huai Hin Lat and Nam Phong formations lie within the confidence interval of the other. A combined mean pole for the Triassic/Jurassic boundary can be derived at $\lambda = 53.6°N$, $\phi = 173.3°E$, $A95 = 4.9°$. When tilting corrections are applied to both the Huai Hin Lat and Nam Phong formations at the site level, the fold test is positive at the 99% level. The positive fold tests and the presence of reversals and of different magnetic carriers (hematite and magnetite) in rocks of various lithologies (limestones and sandstones) suggest a primary nature for the magnetization.

7. Upper Jurassic–lower Cretaceous results

7.1 Sao Khua Formation (upper Jurassic)

Unfortunately, the topography is quite flat in this part of the Khorat basin, west of the Mekong river, so that outcrops are not well exposed; indeed, only a few outcrops of this formation were found in the central and eastern Thai parts of the basin. The dips are low (in general less than 10°), and sometimes difficult to measure with precision. Twelve sites have been sampled, but we have only retained for final computation the ten sites which offer the best structural control. Two components have been isolated using thermal demagnetization (Fig. 7a and b). A low unblocking temperature component is generally removed at temperatures below 300°C. The isolated directions cluster around the direction of the present earth field. A high temperature component with univectorial decay toward the origin is evident in the 500–690°C range (Fig. 7a and b). Saturation

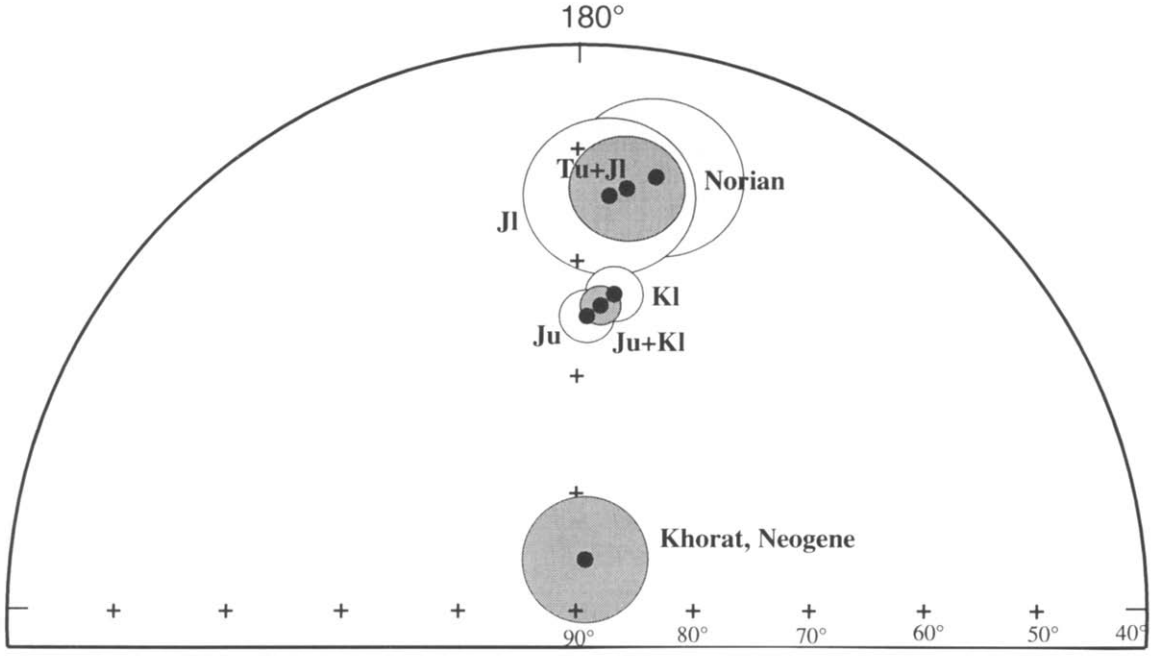
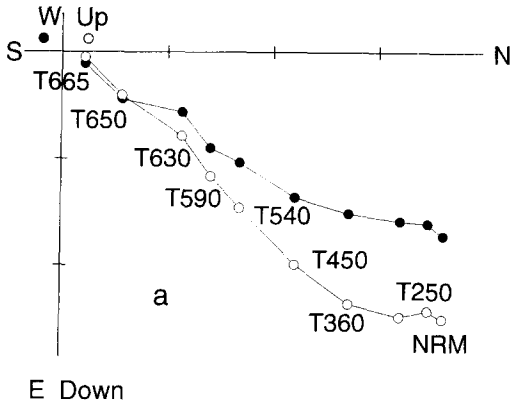
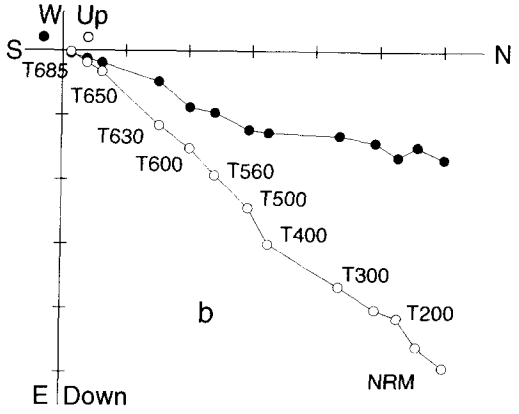


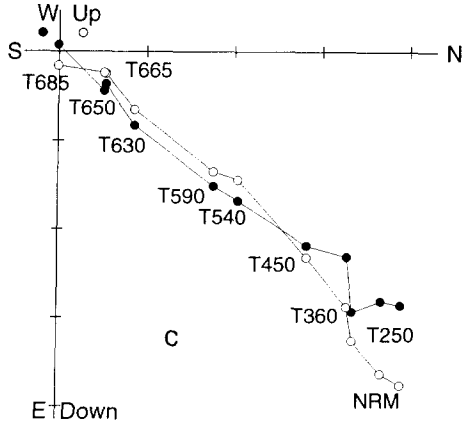
Fig. 6. Upper Triassic to lower Cretaceous pole positions obtained in this study shown on an equal-area projection. A Neogene pole from the Khorat basin [after [45] is also plotted.



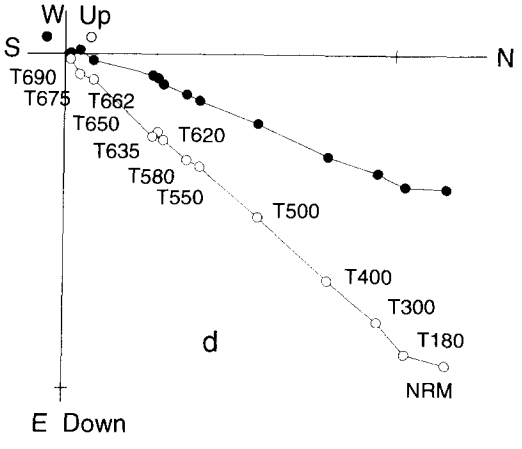
Th 19-8B Scale: 1e-3 A/m



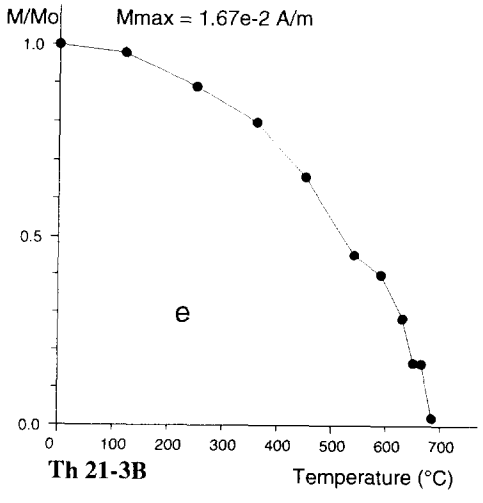
Th 26-6B Scale: 1e-3 A/m



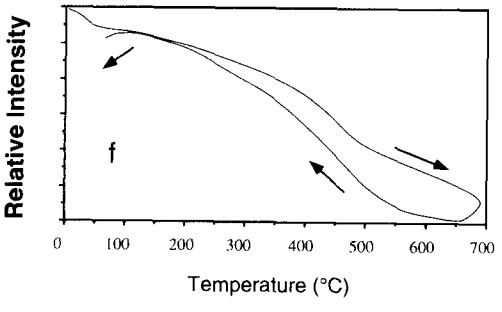
Th 37-5B Scale: 1e-3 A/m



Th 28-8D Scale: 1e-2 A/m



Th 21-3B



Th21-5b

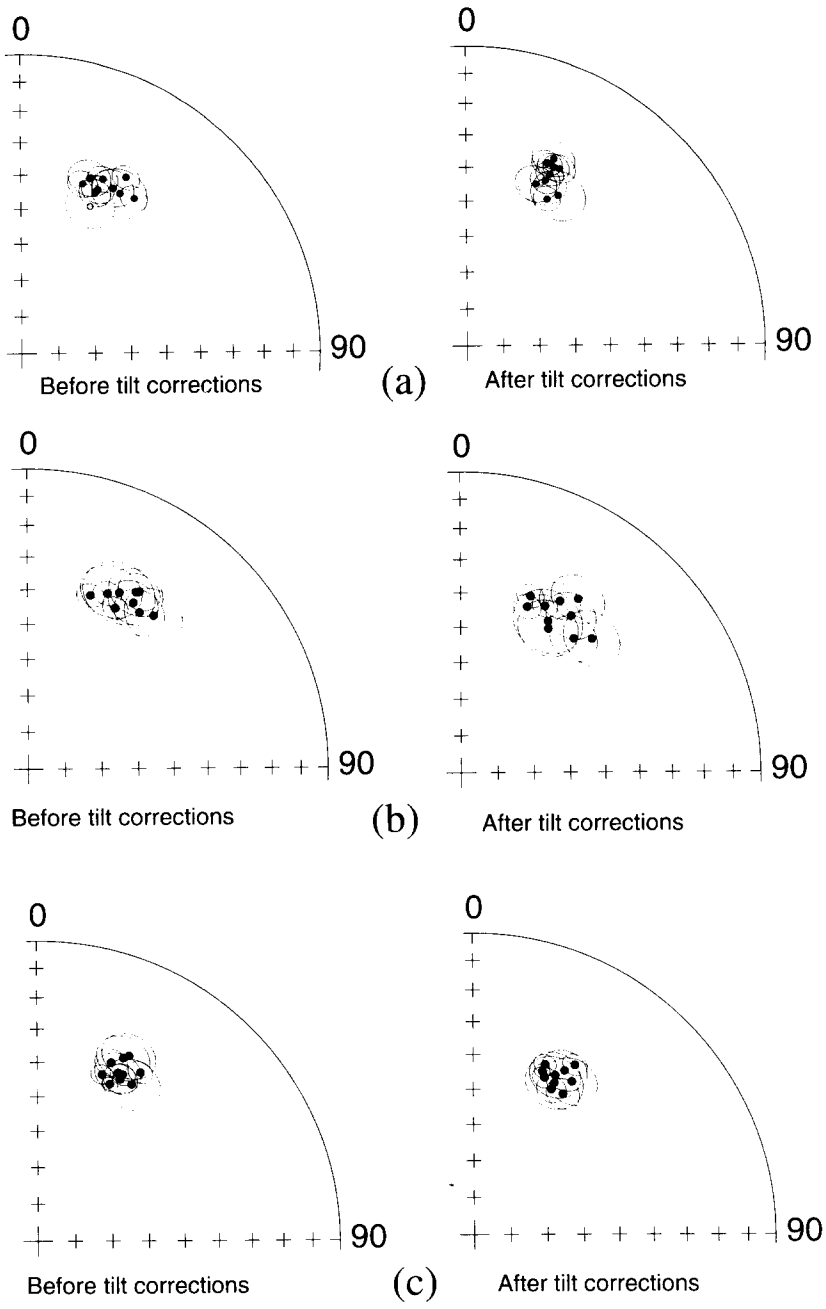


Fig. 8. (a) Equal-area projection for (i) upper Jurassic site-mean directions with their circles of 95% confidence (using a dashed line, the antiparallel direction is also indicated for reversed site 20) and for (ii) the lower Cretaceous for the hematite carrier directions (b) and the magnetite carrier directions (c).

Fig. 7. (a and b) Orthogonal vector diagrams for representative specimens from the Sao Khua Formation (upper Jurassic). (c and d) Orthogonal vector diagrams for representative specimens from the Khok Kruat Formation (Aptian–Albian) during thermal demagnetization, in stratigraphic coordinates. (e) Normalized decay curve of the magnetization intensity during heating. (f) Thermomagnetic curve for the lower Cretaceous samples (site Th21), showing the Curie point of (titano-)magnetite around 500°C and that of hematite at 680°C.

is not reached at 1.3 T and thermal demagnetization of IRM [43] shows a preponderance of a high coercivity mineral with a magnetization loss at 690°C. A medium coercivity component is also evident from thermal demagnetization of the IRM. These characteristics indicate hematite as a possible carrier of the remanence. Curious behaviour is found at site Th20, where a reversed direction emerged at very high temperature, between 670 and 680°C. This is similar to the finding of Huang and Opdyke [44] in some redbeds of the South China block. The magnetic spectra of the normal and reversed components overlap nearly completely; the chemical remanent magnetization probably carried by pigmentary hematite may have formed over a period of time that was long enough to span one (or more) reversal. The direction of this last component is determined using sector-constrained great circle intersections [36] and appears to be antipodal to the directions of normal polarity determined at other sites of the same age from the Khorat basin. When structural corrections are applied, a slightly better grouping of the directions occurs. The mean directions before and after bedding correction differ by only 2°. The mean directions of the ten sites before and after bedding corrections are summarized in Table 3 and are shown in the equal-area projections of Fig. 8a.

7.2 Khok Kruat Formation (lower Cretaceous)

Ten Cretaceous sites of Aptian–Albian age have been sampled (Fig. 1b). The directions of a low temperature component determined before 300°C cluster around the direction of present earth field; two or three components are evident for most of the specimens. All of them are of normal polarity, which is not inconsistent with the fact that the age of the sedimentary rocks corresponds to the long normal superchron. Normalized NRM heat–decay curves (Fig. 7e) first show a medium temperature component with maximum unblocking temperature near 550°C (the remaining intensity being 30–40% of the initial value) followed by a high temperature component destroyed in the range 620 to 670–690°C. Several sites also display a third plateau above 670°C, followed by a rapid decrease in the magnetization. The analysis of Zijderveld plots shows no major change in the direction of the magnetic vector between these components (Fig. 7c and d). However, in some cases, the last points (after 650°C) are rather affected by noise and miss the origin. Occasionally, they also lie in a direction not significantly different from the present earth field. The unblocking temperature range of the last two plateaus (650 and 680°C) leaves little doubt as to the nature of the magnetic carriers, which are, respectively, hematite and pigmentary hematite. The nature of the magnetic carriers for the medium range temperature component requires closer inspection, however. Examination of thin sections using an SEM reveals rounded iron oxide minerals with typical spinel structure and exsolution features suggesting the presence of magnetite. The energy-dispersive spectroscopic spectrum for the large grains of magnetite indicate that the ratio of Ti to Fe is about 1:80. We also examined extracts using X-ray diffraction: The X-ray diffraction pattern for the concentrated magnetic extract of sample Th21-3-5 clearly shows an important amount of hematite, and also some remaining quartz and feldspars. The classical main peak for detrital hematite is observed around 26.6°, along with a large number of well-determined secondary peaks. Magnetite is more difficult to determine as its 100% peak overlaps with the 70% peak of hematite. The possible presence of magnetite is only suggested by the occurrence of the 74% peak. The Curie temperature shows a clear inflexion just above 500°C, while magnetization is completely lost just above 670°C (Fig. 7f). Such Curie temperatures suggest the presence of both Ti-poor titanomagnetite and hematite. Hematite of diagenetic origin could have formed and locked in magnetization a short time after titanomagnetite of possible detrital origin. Two separate means have therefore been calculated (Table 3): the first one uses the medium temperature range directions (magnetite), the other is based on high temperature range directions (hematite). When bedding is corrected, there is a clear increase in k for the medium temperature component, whereas a decrease occurs for the high temperature component (Fig. 8b and c, Table 3). However, the fold test in both cases remains undetermined due to the low dip. The directions of both components are quite similar (Fig. 8b and c, Table 3). Due to the likelihood of

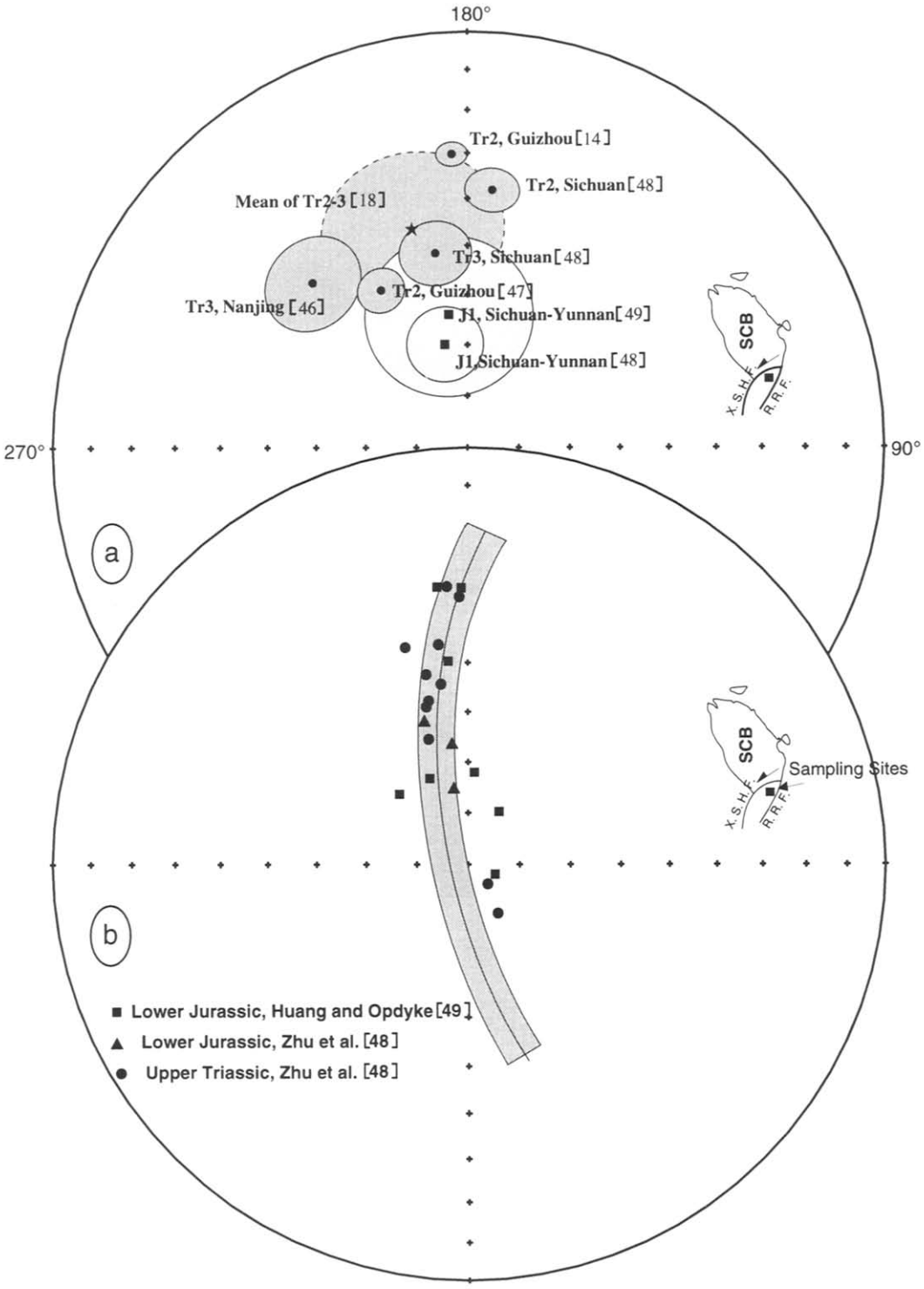


Fig. 9. (a) Early Mesozoic poles from the South China block. The poles are rather scattered in the middle to late Triassic period, yielding a mean pole with a 95% half-angle of confidence as high as 17° (star and dashed line). The scatter in these poles may have resulted from local rotations. (b) The VGPs of the central Yunnan basin plotted on this diagram by site are distributed along a small circle centred on the sampling region.

a slight contamination of the high temperature component, we have used the medium temperature range as the characteristic direction (Fig. 8c). The derived pole appears to have a direction that is very close to that of upper Jurassic age (Fig. 6). A mean Jurassic/Cretaceous boundary pole can be computed at $\lambda = 63.8°N$, $\phi = 175.6°E$, $A_{95} = 1.7°$, for $N = 20$ sites (Table 3).

8. Discussion

A negative fold test was obtained for the Permian succession. The Permian limestones are sometimes recrystallized and are unconformably overlain by the Huai Hin Lat limestone, which is of Norian age, corresponding to the pre-late Triassic Indosinian orogeny of this region. This strongly suggests that the magnetization of the Permian limestones in the Chum–Phae area was acquired during early to middle Triassic folding and thus partly confirms the widespread remagnetization suggested by Chen and Courtillot [24].

A characteristic direction is isolated from the upper Triassic–lower Jurassic Huai Hin Lat and Nam-Phong formations. Two antipodal directions of normal and reversed polarity and a positive fold test at the 99% level are found. Remanence is carried both by magnetite and hematite. These characteristics strongly suggest a primary origin for the magnetization. In particular, the clear message carried by the lower Jurassic Nam Phong Formation (Fig. 5) allows us to finally settle the question of the primary nature of the magnetization of the underlying Norian Huai Hin Lat Formation that could not be solved with the data available to Achache and Courtillot [23] and Chen and Courtillot [24]. The lack of a significant fold test at the 95% level (there is just a small increase in the precision parameter) does not allow us to ascertain the primary nature of the upper Jurassic–lower Cretaceous magnetization. However, one can see that the mean direction is significantly different from that of the Triassic–Jurassic field (Fig. 6). The lower Cretaceous and upper Jurassic successions that we have sampled are located directly above the upper Triassic–lower Jurassic succession for which a positive fold test is available. It is difficult to imagine a remagnetization of the upper Mesozoic samples which would have left the lower Mesozoic succession, which is located immediately beneath, untouched. Moreover, the directions from which this pole has been computed are significantly different from the present-day field, and from the Neogene direction of McCabe et al. [45], immediately to the south of the sampled area (Fig. 6). We thus feel that the upper Jurassic–lower Cretaceous magnetization is also of primary origin, or at least was acquired shortly after the deposition of the sediments. The two pole positions imply an absolute clockwise rotation of the Indochina block at a nearly constant latitude, somewhat to the north of its present-day position.

8.1 Paleoposition of the Khorat plateau relative to the South China block

An exhaustive list of Asian poles has recently been published by Enkin et al. [18]. From their selection of poles they determined APWPs for several southeast and central Asian blocks, including the South China block. The south Chinese APWP appears to be well determined for certain periods, with a large number of poles coming from reliable studies. Such is particularly the case for the upper Permian and the lower Triassic, for which good quality pole determinations, with sites geographically distributed over a wide area of the block, impart some confidence to the corresponding mean poles. Some other periods, such as the lower–middle Jurassic and middle–upper Triassic, lack coherent data (Fig. 9a). A clear example is given by the Tr2–Tr3 and J1 poles of Enkin et al. [18]. These poles are important because they constrain a substantial eastward loop of the APWP (although of course with large, and acknowledged, uncertainty). The consequences of the assemblage of Asian blocks during this key period, and particularly of the collision and suturing histories between the North and South China blocks, are significant, with greater extension of the Song-Pan Ganze region in the late Triassic than in the early Triassic, between the NCB and SCB through western Qinling.

We have tried to re-analyse the data set for this time period. The scatter of available Tr2 and Tr3 poles (Fig. 9a) precludes the computation of any reliable middle or upper Triassic mean pole [18]. Moreover, these poles do not fulfill our

minimum acceptability criteria. For example, the T3S3 pole from Nanjing [46] was determined after the analysis of 23 specimens, while the T2S1 pole from Chan [47] is based upon only eight samples. An upper Triassic pole (T3S4 in Enkin et al. [18], from Zhu et al. [48]) and two poles for the lower Jurassic are of relatively high quality (one pole already used in the Enkin et al. [18] compilation (J1S1 in their pole list, from Zhu et al. [48]) and a new one published recently by Huang and Opdyke [49]). We have plotted the VGPs from each site in these three studies on the equal-area projection of Fig. 9b. This diagram shows a clear elongated trend for the poles, located more or less on a small circle centred on the coordinates of the sampling site. This elongation is clearly not correlated to the age of the sites. All these poles come from the same region (the southwestern part of Sichuan and central Yunnan basin) in an area situated west of the Xian–Shui–He fault system, which is known to have been strongly active during the India–Asia collision [50,51]. It is more than probable that these regions have suffered important tectonic strain resulting in noticeable rotations about vertical axes. This confirms that the mean Tr2–Tr3 and J1 poles published by Enkin et al. [18] and, consequently, the existence of a cusp of upper Triassic–lower Jurassic age in the APWP of South China, are not reliable, as the sites from which the poles were computed were not rigidly attached to the main part of the SCB.

The determination of new middle Triassic to middle Jurassic poles from the South China block is thus urgently needed. Even if the local rotations encountered preclude the determination of a pole, the paleoinclinations can be used to compute the paleolatitudes of the sampled terranes. In order to avoid the bias due to arithmetic averaging of inclinations only, we have used the statistical method of McFadden and Reid [52]. Using the lower Jurassic inclinations [48,49] and those of the upper Triassic [48] we found a paleolatitude of 23.7°N ± 2.5° for a point presently situated at 26°N, 102°E (central Yunnan basin). The possible pole positions based on the inclinations only (Figs. 9b and 10a) appear to be distinct at the 95% level from the Tr3–J1 pole of the Khorat plateau. This therefore shows a significant southward motion of Indochina relative to the SCB. A paleolatitude of 31.6°N ± 5.0° (when reduced at the same reference point from our Triassic–Jurassic pole of the Indochina block) is found for the Khorat basin of the Indochina block, implying a latitude difference of 7.9 ± 5.6° and placing the Khorat plateau in the same band of latitude as the central Yunnan basin during Triassic–Jurassic times, and for obvious topological reasons necessarily to the west of it.

Despite recent efforts, precise determination of poles for the Cretaceous and upper Jurassic of the South China block is not a simple matter either. The first point, and one which has often been emphasized in the literature, is the absence of precise ages due to the continental nature of the formations (primarily red sandstones devoid of good stratigraphic fossils). The above discussion of the Tr3–J1 sites also leads us to exclude J3–K1 poles coming from sites west of the Xian–Shui–He fault. Our determinations of the upper Jurassic–lower Cretaceous pole for the South China block is therefore slightly different from that of Enkin et al. [18]. For the time span considered we have retained the poles K1S3, K1S4, J3S1 and J3S2 from the list of Enkin et al. [18], and have added to this a newly published K1–K2 pole of Hu et al. [53] from Fujian province. As emphasized by Courtillot and Besse [54] and Enkin et al. [55], the upper Jurassic and lower Cretaceous poles of the SCB are within the common intersection of their 95% circle of confidence. An upper Jurassic–lower Cretaceous pole can thus be derived at 74.3°N, 215.7°E ($A95$ = 6.5°) from these five studies (Fig. 10b). Enkin et al. [18] noted that the lower Cretaceous pole from Zhu et al. [48] in central Yunnan was a clear outlier. Recent data from Huang and Opdyke [fig. 3 in 44] also show a spread in the declination values. These sites are in the same structural position as their J1 sites, west of the Xian–Shui–He fault system, and have probably undergone similar local rotations related to the India–Asia collision. We have computed a mean lower Cretaceous–upper Jurassic paleolatitude at 23.6°N ± 2° for these 50 sites, using inclinations only for the Huili area (central Yunnan basin) of the SCB [44,48].

The circular band of possible pole positions inferred from these data has been plotted in Fig. 10b. The pole from Khorat appears to be distinct,

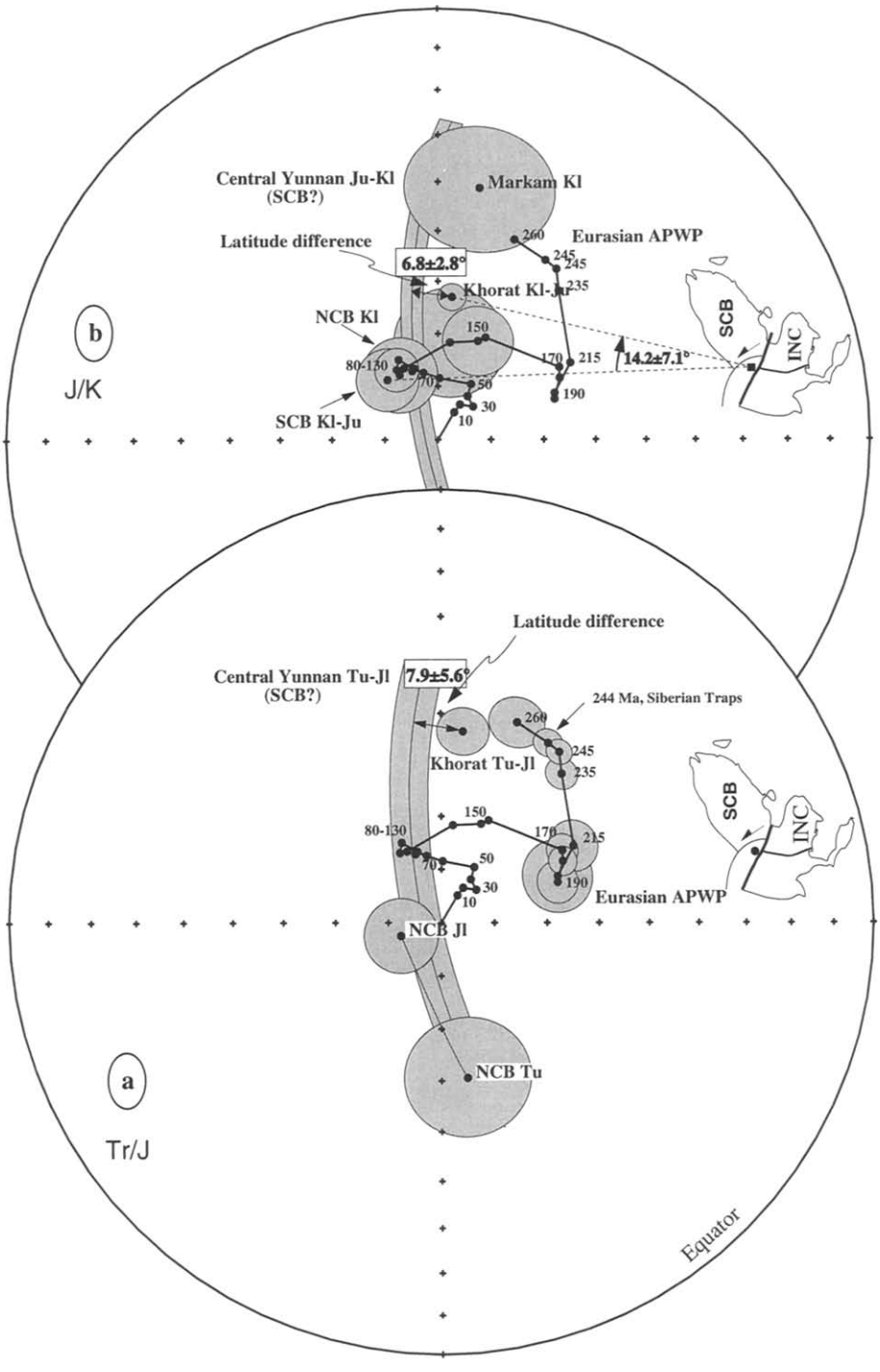


Fig. 10. Paleomagnetic poles discussed in the text for the Triassic–Jurassic (a) and the Jurassic–Cretaceous (b). The APWP for Eurasia is after Besse and Courtillot [60], Van der Voo [56] and Théveniaut et al. [59]. The upper Permian pole from the Siberian Traps is from the Lock and McElhinny database [58]. The possible pole positions (with 95% error bounds) of the Huili area (central Yunnan basin) are also plotted, as a small circle centred on the sampling area (■).

at the 95% level of confidence, from both poles of the SCB and central Yunnan area and rotated clockwise relative to the SCB. The paleolatitude and declinations of the Khorat plateau and of the South China block calculated at our reference site (26°N, 102°E) are found to be, respectively, $\lambda = 30.4°N \pm 2.0°$, $D = 29.4 \pm 2.0°$, and $\lambda = 18.9°N \pm 6.5°$, $D = 15.2 \pm 6.9°$. The latitudinal difference between the Khorat plateau and the central Yunnan basin at that time is $6.8 \pm 2.8°$, which is very close to the $7.9 \pm 5.6°$ found at the Triassic/Jurassic boundary, suggesting that major motion in terms of latitude occurred only after the end of the Mesozoic. Local rotations preclude the determination of the relative rotation with respect to the central Yunnan basin. Alternatively, the total amount of latitudinal displacement between the Khorat and the South China block is $11.5 \pm 6.7°$ and the total rotation $14.2 \pm 7.1°$ at our reference site (Fig. 10b). Finally, there is a small, insignificant, difference in paleolatitude of $4.7 \pm 6.8°$ between central Yunnan and the South China block (Fig. 10b).

We have plotted in Fig. 10 the APWP for Eurasia for further discussion. From 260 to 215 My we have used the synthesis of Van der Voo [56]. Two poles have been added. One is a mean pole derived from 102 studies of the Siberian traps, recently redated at 244 ± 1 Ma by Baksi and Farrar [57]. This pole, obtained from a compilation of the Lock and McElhinny paleomagnetic database [58], is in excellent agreement (Fig. 10a) with that derived by Van der Voo [56] from a compilation of European poles and of North American poles transferred onto Eurasia using the kinematics of the opening of the North Atlantic Ocean. The direct consequence is that poles from Western Europe can probably be used safely to assess the paleoposition of stable Siberia since at least the Permo-Triassic boundary. We have also used a pole from Théveniaut et al. [59], which is the only pole available for the middle Triassic of Europe. Finally, the segment between 190 My and the present is the synthesis APWP of Besse and Courtillot [60]. We have also plotted the segment of APWP for the upper Triassic, lower Jurassic and lower Cretaceous poles for the North China block (NCB) [17,18,61]. It appears that, during the Triassic/Jurassic boundary, the NCB, central Yunnan and Indochina have poles streaked along small circles more or less centred on present-day southeast Asia. These poles are remote from Eurasian poles of the same age. The configuration is almost the same at the Jurassic/Cretaceous boundary, the pole from Khorat being situated (within uncertainty limits) at the same latitudinal position relative to central Yunnan (the SCB confidence interval intersecting the path of possible pole positions of Yunnan) while the Chinese blocks were assembled almost in their final positions relative to Eurasia. These data suggest that NCB, SCB, Yunnan and Indochina were probably in contact with each other since at least the late Triassic [see also 17,18] and moved slowly northward relative to Eurasia until the Cretaceous. The details of the collision between the SCB and NCB cannot be discussed further because no reliable data constrain the position of the SCB during the middle–late Triassic to early Jurassic. The similarity of the relative pole positions of Khorat and Yunnan shows that these two blocks underwent no important motion in terms of latitude during the late Triassic to early Cretaceous, whereas important post-Cretaceous displacement of Khorat relative to both the central Yunnan basin and the SCB is implied.

8.2 The extrusion of Indochina by continental collision

Since the collision of India with Eurasia at about 50 My roughly N–S intracontinental convergence has occurred at a rate of around 5 cm/yr. A combined analysis of magnetic anomalies from the Indian Ocean [8] with paleomagnetic data on the northern edge of India and the southern edge of Eurasia [7,20] has shown a N–S convergence in excess of 2600 km. This shortening is made up of 1900 ± 850 km between stable Eurasia and the Lhasa block (the former southern margin of Eurasia) and an additional 700 ± 300 km of convergence is found between stable India and the Lhasa block. Recent paleomagnetic studies on Cretaceous rocks from Tarim [16,19], the Junggar block [21] and Tibet [19,20,62] indicate distributed deformation within central Asia and confirm the total amplitude of convergence. Large crustal shortening between India and Siberia is also absorbed within the Tien Shan and the Altay foldbelts.

Several models have been proposed to accommodate the northward motion of India. The first involves underthrusting of India beneath Eurasia [63,64]. However, as we recalled above, paleomagnetic study of the Cretaceous paleomargin of Eurasia indicates that a large part of the deformation occurred within Eurasia itself. Two other models are also proposed—crustal shortening and thickening only [65–68], and lateral extrusion of continental blocks along large-scale strike-slip faults [10,13,69–72]. The predictions of these models are at variance regarding the amount and mode of shortening since the onset of the collision. According to the first group of authors, crustal shortening and thickening are sufficient to accommodate the convergence. In this case, second-order latitudinal movements and important rotations between the blocks involved should be observed; in the latter case, both significant rotations and N–S translations should be found by paleomagnetic study.

A simple reconstruction at the Jurassic/Cretaceous boundary is proposed in Fig. 11. We have characterized the main blocks in the following way:

The South China block is bounded to the north by the Qinling mountain range, to the west by the Long Men Shan, a large-scale Tertiary thrust belt, and to the southwest by the Xian–Shui–He and Red River faults. These two faults have been studied extensively [50,51,72,73]. The Xian–Shui–He fault, which is one of the most active strike-slip faults in the world, can be observed along a distance in excess of 1000 km in the Song Pan accretionary complex [50,51,74]. It is split into two branches, which traverse the central Yunnan basin before disappearing into a more diffuse folded deformation zone to the

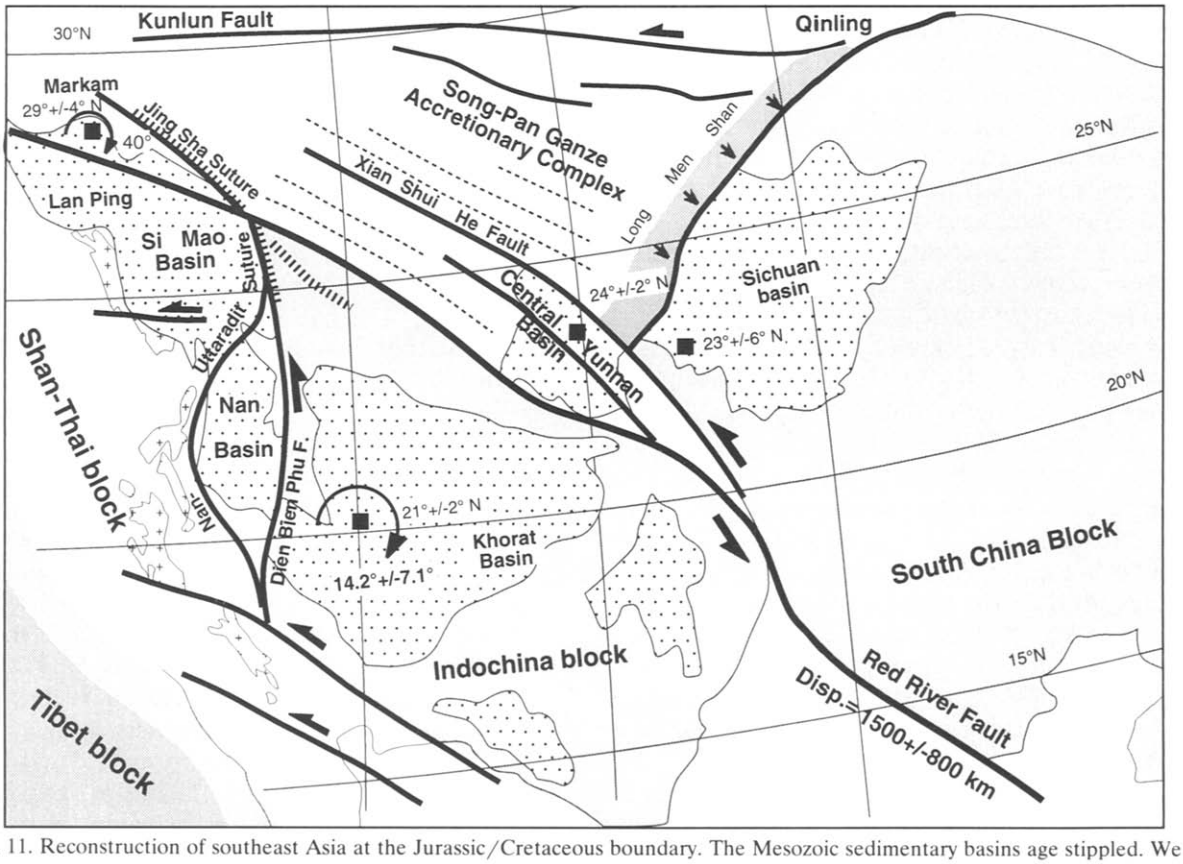


Fig. 11. Reconstruction of southeast Asia at the Jurassic/Cretaceous boundary. The Mesozoic sedimentary basins age stippled. We have tentatively drawn (heavy lines) the main faults that were initiated or reactivated during the Indian–Asian collision. The black arrows indicate the main sense of motion on these faults. The Long Men Shan thrust zone is also shown. Axes of the main folds are shown as short-dash lines [after 51]. The motion inferred on the Red River fault reaches 1500 ± 800 km. Rotations are indicated relative to the South China block.

south, on the Yangtze platform. The motion inferred from field data is clearly sinistral throughout its entire history [51]. The Red River fault is another major strike-slip fault, also exceeding 1000 km in length, which extends between the Gulf of Tonkin and Tibet. Recent fieldwork and U/Pb dating in the Ailao Shan [72,75] demonstrate sinistral motion on this fault at least prior to 23 My, although recent motion on parts of the fault has involved both normal and right-lateral slip [72,73].

The Khorat plateau (Indochina block) is bounded by the Red River fault zone to the northeast and by the Dien Bien Phu fault to the west, more or less following the ancient Nan–Utarradit suture zone which may mark the boundary between the Shan–Thai Malay block and the Indochina block [1]. The southern edge of the Indochina block corresponds to the Wang Chao and Three Pagodas sinistral fault zones [10].

We have drawn the present-day coast in Fig. 11 for more clarity, together with the outlines of basins such as the Khorat, Sichuan, central Yunnan, Si-Mao, Nan and Markam. Most of these basins share common features, consisting in upper Permian marine sediments on which a thick sequence of continental sediments of Triassic to upper Cretaceous (or Paleocene?) age is deposited. In some places, lacunas or unconformities of the upper Triassic sediments over older rocks mark the Indosinian orogeny. These may have been created by the probable collision of the Qiangtang block against the Eurasian margin and the clamping of the Song-Pan Ganze during the northward motion of both Indochina and the South China block [4,74,76]. We have maintained the present shape of the blocks (i.e., we have ignored internal deformation). The main blocks have been positioned according to their most probable paleolatitude as deduced from the paleomagnetic studies, and their azimuth was restored using paleodeclinations. The western part of the Yunnan basin has been restored using paleomagnetic data for the Huili area from Huang and Opdyke [44]. We note that despite the large spread in declination and the resulting uncertainties, several directions have clearly been rotated strongly clockwise. The paleomagnetic data of Otofuji et al. [77] and Huang et al. [78] from the Qamdo–Markam area (Fig. 10b) have also been used to restore the paleoposition of this region. The latitudes computed from these data are very close to the present-day position. Due to the large spread in declination values, we have recomputed a mean paleolatitude using only inclinations from [77] and [78]. A paleolatitude of 30.1° is derived, with a smaller error of 4.0°. Recent studies on the Cretaceous sedimentary rocks of South Tibet [20,62,79,80] and on sedimentary rocks in western North Tibet [19] show that the southern margin of Tibet (that is to say the southern margin of Eurasia at that time) was situated at a latitude close to 10–15° N. This would place the North and South Tibet blocks immediately to the southwest of the Shan–Thai–Malay and Indochina blocks, in agreement with tectonostratigraphic correlations [81]. From Fig. 11 we can observe that several basins were formed in tropical latitudes over a wide area during the Mesozoic.

The latitudinal difference between the Khorat and Sichuan basins demonstrated by paleomagnetism implies significant movement on the Red River fault. By taking a paleoazimuth of about N35°W for this fault, sinistral motion of 1500 ± 800 km can be deduced for the period since the late Jurassic to early Cretaceous. The relative configuration of landmasses at this time is unlikely to have differed significantly from that prevailing shortly after the Cretaceous–Tertiary boundary, prior to the onset of the India–Asia collision. As a result, this displacement value may be close to the one which can be directly related to the Tertiary deformation of Asia during the penetration by India. This value is on the high side of that predicted by the extrusion model of Tapponnier et al. [10] based on the lateral offset of several geological markers, and of that predicted by the kinematic model of Briais et al. [82] based on the analysis of the South China Sea magnetic anomalies (an offset of between 500 and 600 km, which is our lower bound at the 95% confidence level). The age of the opening of the South China Sea is thought to lie in the 32–17 My range [72,82]. However, the amount of motion on the Red River fault computed following this kinematic model includes only oceanic seafloor, to which a pre-opening extension of some 100–200 km taking place within the margins could also be

added. Moreover, the amount of right-lateral displacement on the Red River fault since the Pliocene [72,73] could be as much as 100 km [P. Tapponnier, pers. commun., 1991]. A total sinistral displacement of 700–900 km on the Red River fault is therefore likely. The rotation of the Khorat relative to the South China block is clockwise, with a value of $14.2 \pm 7.1°$. This value is very similar to that predicted by the kinematic model of Briais et al. [82]. North-south displacements of $4.7 \pm 6.8°$ of Yunnan in the Huili area and of $8.4 \pm 7.6°$ in the Markam Qamdo area relative to the stable South China block are also found. Although these values of displacement are not significant at the 95% level, the most probable values of paleolatitude indicate that several small blocks may have also been extruded along left-lateral strike-slip faults, possibly between the Xian–Shui–He fault zone and the Red River faults or any of their predecessors. These smaller blocks all display strong clockwise rotation values, in agreement with the extrusion model [13]. Our reconstruction places the remnants of the Triassic Jinsha suture zone, directly to the east of Markam, as a direct continuation of the Nan–Uttaradit suture zone, corresponding to 600 km of sinistral motion on the Red River fault, much to the west of where it joins the Xian–Shui–He fault. This thus restores the ancient probable continuity between these two fragments of an ancient suture of late Triassic age and implies that several hundred kilometres of strike-slip must indeed have taken place on fault systems other than the Red River, branches of the Xian–Shui–He being the most prominent candidates. This study fully supports the key roles of extrusion as an important mode of crustal shortening during the India–Asia collision. To the south, close to the Gulf of Tonkin, the motion is accommodated within a narrow area concentrated along the Red River fault. Further north, the deformation is distributed over a wider area in which several faults take up components of the total movement. The Song Pan Ganze Triassic chain was thus strongly reactivated during the Tertiary [51].

9. Conclusion

This study partly confirms the remagnetization suggested by Chen and Courtillot [24]. This remagnetization event, however, was probably related to the early or middle Triassic Indosinian orogeny. Two paleomagnetic poles, of upper Triassic–lower Jurassic and upper Jurassic–lower Cretaceous age, have been derived and show that Indochina underwent a clockwise rotation at a nearly constant tropical latitude during this period. The directions of upper Jurassic to Aptian–Albian age are nearly identical, and can be compared with the well-known standstill of North and South China that has already been noted by several other authors [e.g., 55] for this period.

Careful analysis of the APWP for the South China block has shown that regions (central Yunnan) located to the west of the Long–Men–Shan thrust and Xian–Shui–He strike-slip fault zones do not represent the stable SCB. These zones have suffered significant strain, which is marked by strong local rotations. No reliable paleomagnetic data constrain the paleoposition of the stable SCB from the middle Triassic to the middle Jurassic. However, the comparison of available directions from Indochina, SCB, NCB and Yunnan suggests that these blocks were in contact with each other at least since the late Triassic, even if they may have been in significantly different relative positions [e.g., 18,37].

A comparison of our paleomagnetic directions for Indochina with those of adjacent blocks situated to the east, at the Jurassic/ Cretaceous boundary, shows the following:

(1) Sinistral motion on the Red River fault system of 1500 ± 800 km since this period. The configuration of the blocks at this time was probably the same as that prior to the India–Asia collision, so that this motion most probably derives from strain linked to the collision. This estimate of the sinistral motion is larger than that of Tapponnier et al. [10,72] and Briais et al. [82], yet, given the rather large uncertainties, not incompatible.

(2) A clockwise rotation of $14 \pm 7°$ of Indochina relative to the SCB is found, in agreement with the estimates given by Tapponnier et al. [10,72] and Briais et al. [82].

(3) Blocks situated west of the stable SCB (e.g., the Markam area, Song-Pan Ganze and central Yunnan basin) were also probably extruded as fairly narrow strips along with the main body of Indochina. They all display strong clock-

wise rotations, again in agreement with the extrusion model of Peltzer and Tapponnier [13].

Finally, this study shows that extrusion is one of the main mechanisms that accommodates crustal shortening during the India–Asia collision.

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